

CSCI3230 (ESTR3108)

Fundamentals of Artificial Intelligence

Lecture 2. Linear Regression

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Outline

Part 1. Linear regression problem

Part 2. Optimizing linear regression

Part 3. Shrinkage methods

Part 4. Summary



Part 1. Linear regression problem

Concepts of regression

What is regression?

¹https://en.wikipedia.org/wiki/Regression_analysis

Concepts of regression

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- Regression is a statistical processes for estimating the relation between dependent variables (often called “outcome” or “response variable”) and independent variables (often called “predictor”, “covariates”, “explanatory variables” or “features”).¹

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- The input of regression program is independent variable(s).
- The output of regression program is dependent variable(s).

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Concepts of linear regression

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- Intuitively, graphs of linear functions are straight lines / planes.
- Mathematically, a function $f(x)$ is linear iff $f(u + v) = f(u) + f(v)$ and $f(cu) = cf(u)$.

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Why linear regression?

- Linear relationship is very simple yet useful to model the world. Linear relationships are pretty common in daily life, e.g., kinetic friction and normal force, gravity and mass, distance and speed, etc.

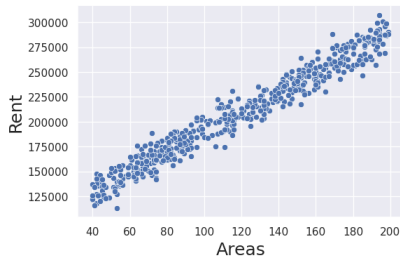
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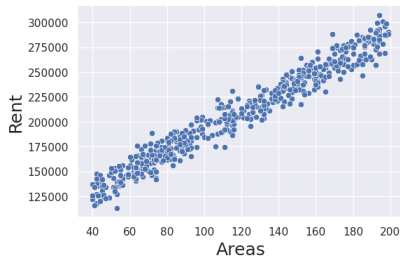
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1	40	134072
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...	...	...



A concrete example

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- Suppose we already have a dataset like the table below and plot the data points in the right figure.
- Can we predict the rent of any input apartment areas?

Index	Areas (m^2)	Rent (HKD)
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Basic formulation

Consider intelligent machines as functions with input and output

- $f : X \rightarrow Y$, where $X \subset \mathbb{R}^n$ is the domain of input and $Y \subset \mathbb{R}$ is the domain of output.

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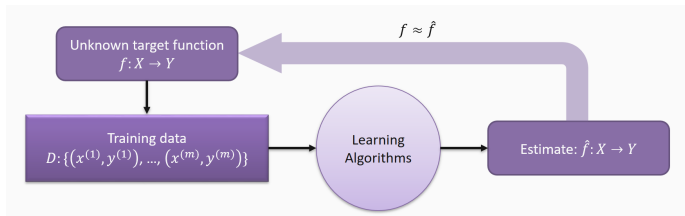
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- In the example of “rent and areas”, X is the domain of apartment areas and Y is the domain of rent.

Basic formulation

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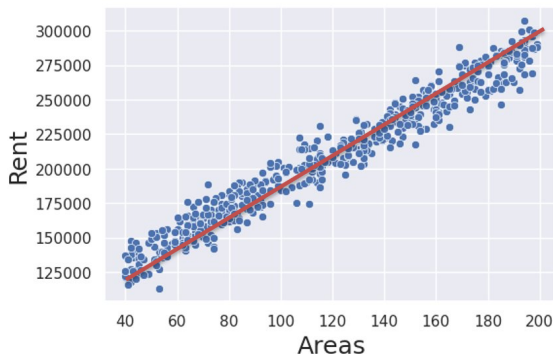
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- In the example of “rent and areas”, X is the domain of apartment areas and Y is the domain of rent.

Goal: take the given data as training examples and try to find a general mapping $\hat{f} : X \rightarrow Y$, which is close to true f as much as possible.



Formulation for 1-dimensional feature space

- Observe the plot of data points, it seems these data points are distributed along a straight line.
- For simplification, we can assume the relation between input X and output Y is **linear**.



Formulation for 1-dimensional feature space

Univariate Linear Function

$$\hat{y} = \hat{f}_{\theta_0, \theta_1}(x) = \theta_1 x + \theta_0$$

Formulation for 1-dimensional feature space

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If we manage to find “the best” θ_1 and θ_0 , namely the relation between apartment areas and rent is determined, it is possible to predict the rent by given any apartment areas.

Formulation for n-dimensional feature space

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What if the rent is also related to other features (factors), e.g., the distance to railway station and floors?

- We need to bring in n variables $x_1, x_2, \dots, x_n$ to represent n features.
- In this way, we utilize a linear function with n variables for n -dimensional feature space:

Multivariate Linear Function

$$\hat{y} = \hat{f}_{\theta_0, \theta_1, \dots, \theta_n}(x_1, \dots, x_n) = \theta_0 + \theta_1 x_1 + \dots + \theta_n x_n$$

Formulation for n-dimensional feature space

Multivariate Linear Function

Write the multivariate linear function in matrix form:

$$\hat{y} = \hat{f}_{\Theta, \theta_0}(X) = X^T \Theta + \theta_0$$

where $\Theta = \begin{pmatrix} \theta_1 \\ \vdots \\ \theta_n \end{pmatrix}$, and $X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$.

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Please keep in mind θ_0 and Θ are the parameters that should be **learned** from training data.

Formulation for n-dimensional feature space

How about representing a linear regression problem with m samples?

- Suppose we have

$$X^{(1)} = \begin{pmatrix} x_1^{(1)} \\ \vdots \\ x_n^{(1)} \end{pmatrix}, X^{(2)} = \begin{pmatrix} x_1^{(2)} \\ \vdots \\ x_n^{(2)} \end{pmatrix}, \dots, X^{(m)} = \begin{pmatrix} x_1^{(m)} \\ \vdots \\ x_n^{(m)} \end{pmatrix}, \text{ with}$$

respective labels $y^{(1)}, y^{(2)}, \dots, y^{(m)}$.

- We denote $\mathbf{X} \in \mathbb{R}^{m \times n}$ as the data matrix, of which rows represent samples, columns represent features.

$$\mathbf{X} = \begin{pmatrix} X^{(1)T} \\ \vdots \\ X^{(m)T} \end{pmatrix} = \begin{pmatrix} x_1^{(1)} & \dots & x_n^{(1)} \\ \vdots & \ddots & \vdots \\ x_1^{(m)} & \dots & x_n^{(m)} \end{pmatrix}$$

Formulation for n-dimensional feature space

Matrix Representation of Linear Function

The function for linear function with n features and m samples could be written as

$$\hat{Y} = \hat{f}_{\Theta, \theta_0}(\mathbf{X}) = \mathbf{X}\Theta + \theta_0$$

where $\Theta = \begin{pmatrix} \theta_1 \\ \vdots \\ \theta_n \end{pmatrix}$, $\hat{Y}$ is in the form of $\begin{pmatrix} \hat{y}^{(1)} \\ \vdots \\ \hat{y}^{(m)} \end{pmatrix}$.

Formulation for n-dimensional feature space

Please pay attention to the interpretation of the matrix representation:

$$\mathbf{X}\Theta = \theta_1 \begin{pmatrix} x_1^{(1)} \\ \vdots \\ x_1^{(m)} \end{pmatrix} + \theta_2 \begin{pmatrix} x_2^{(1)} \\ \vdots \\ x_2^{(m)} \end{pmatrix} + \cdots + \theta_n \begin{pmatrix} x_n^{(1)} \\ \vdots \\ x_n^{(m)} \end{pmatrix}$$

You can understand it as mixing features for each sample separately.

Formulation for n-dimensional feature space

- For simplification, usually absorb the bias term θ_0 to the Θ .
- Then insert a column filled with ones to $\mathbf{X}$.

$$\mathbf{X} = \begin{pmatrix} \mathbf{1} & x_1^{(1)} & \dots & x_n^{(1)} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{1} & x_1^{(m)} & \dots & x_n^{(m)} \end{pmatrix}, \quad \Theta = \begin{pmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_n \end{pmatrix}$$

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- Finally, we reach the simplified version:

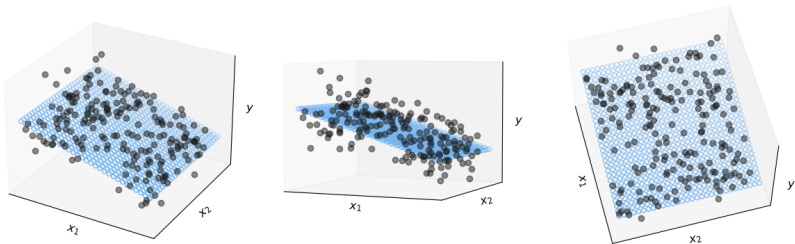
Simplified Matrix Representation of Linear Function

$$\hat{Y} = \hat{f}_{\Theta}(\mathbf{X}) = \mathbf{X}\Theta$$

which will make the subsequent derivation much more handy.

Geometry of linear regression

- Data points $\{(x_1^{(1)}, \dots, x_n^{(1)}, y^{(1)}), \dots, (x_1^{(m)}, \dots, x_n^{(m)}, y^{(m)})\}$ occupies a $(n + 1)$ -dimensional space.
- The “fitting lines” for high-dimensional feature space are n -dimensional **hyperplanes**.
- Examples for linear regression on 2-dimensional feature space:





Part 2. Optimizing linear regression

Problem settings

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- Since all data points are supposed to be sampled from the same distribution, hopefully, once a Θ^* “perfectly” fits the training data, $\hat{f}_{\Theta}$ should be very close to the real f .

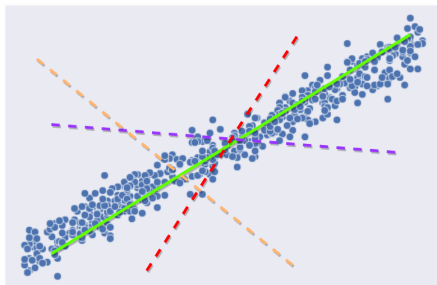
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- Thus we firstly focus on fitting the training data.

Problem settings

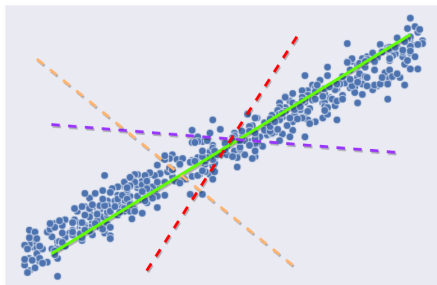
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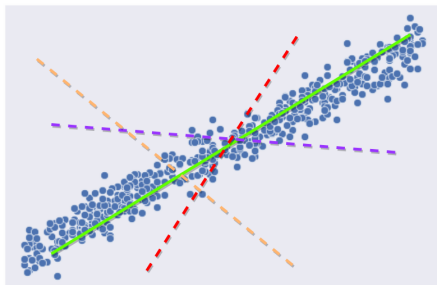
- Intuitively, we need to measure the distance between the estimated values $\hat{f}(X_i)$ and the training data true values (also called labels).



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- Intuitively, we need to measure the distance between the estimated values $\hat{f}(X_i)$ and the training data true values (also called labels).
- This kind of distance should be a function of Θ .



Objective function

- We denote $J(\Theta)$ as the **objective function** (a.k.a. **cost function**) to measure the distance.

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- **Goal:** find the Θ^* that minimizes the function $J(\Theta)$.

Residual Sum of Squares

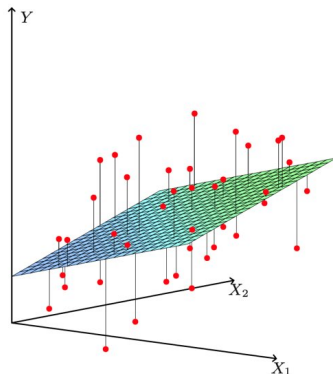
In general, we define $J(\Theta)$ as **Residual Sum of Squares (RSS)**:

$$J(\Theta) = \sum_{i=1}^m (\hat{f}_{\Theta}(X^{(i)}) - y^{(i)})^2 = \|\hat{f}_{\Theta}(\mathbf{X}) - Y\|_2^2$$

The method that approximates the solution by minimizing the RSS is called **ordinary least squares**.

Geometry of RSS

For each sample, “residual” means the difference between the estimated value (the plane) and the corresponding training label (red points).



Example of solving univariate linear regression

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- RSS for univariate linear regression:

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- After expanding the quadratic term and reduction, we have:

$$\begin{aligned} J(\theta_0, \theta_1) = & \sum_{i=1}^m (x^{(i)} - \bar{x})^2 \left(\theta_1 - \frac{\sum_{i=1}^m (x^{(i)} - \bar{x})(y^{(i)} - \bar{y})}{\sum_{i=1}^m (x^{(i)} - \bar{x})^2} \right)^2 + m(\theta_0 - (\bar{y} - \theta_1 \bar{x}))^2 \\ & + \sum_{i=1}^m (y^{(i)} - \bar{y})^2 - \frac{(\sum_{i=1}^m (x^{(i)} - \bar{x})(y^{(i)} - \bar{y}))^2}{\sum_{i=1}^m (x^{(i)} - \bar{x})^2} \end{aligned}$$

where $\bar{x}$ and $\bar{y}$ are means of $x^{(i)}$'s and $y^{(i)}$'s, respectively.

Example of solving univariate linear regression

- Note that only the first two terms are affected by θ_0 and θ_1 .
- To minimize $J(\theta_0, \theta_1)$, let the first two terms equal to 0 (why?).

$$\begin{cases} 0 = \sum_{i=1}^m (x^{(i)} - \bar{x})^2 \left(\theta_1 - \frac{\sum_{i=1}^m (x^{(i)} - \bar{x})(y^{(i)} - \bar{y})}{\sum_{i=1}^m (x^{(i)} - \bar{x})^2} \right)^2 \\ 0 = m(\theta_0 - (\bar{y} - \theta_1 \bar{x}))^2 \end{cases}$$

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$$\implies \begin{cases} \theta_1 = \frac{\sum_{i=1}^m (x^{(i)} - \bar{x})(y^{(i)} - \bar{y})}{\sum_{i=1}^m (x^{(i)} - \bar{x})^2} \\ \theta_0 = \bar{y} - \theta_1 \bar{x} \end{cases}$$

Quiz

Please use least squares to find the line regression solution for

$$y = \theta_1 x + \theta_0$$

with the training data points represented in (x, y) pairs as below.

$$(9, 21), (3, 6), (12, 27)$$

- A $\theta_1 = 2, \theta_0 = 1$
- B $\theta_1 = 2.3571, \theta_0 = -0.8571$
- C $\theta_1 = 2.1456, \theta_0 = -1.7384$
- D $\theta_1 = -2.5078, \theta_0 = 1.0878$

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Correct Answer: B

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- We formulate the ordinary linear regression as

$$\hat{\Theta} = \Theta^* = \operatorname{argmin}_{\Theta} J(\Theta)$$

Convex objective function

Let's go back a bit and think about why we choose RSS as the objective function?

Convex objective function

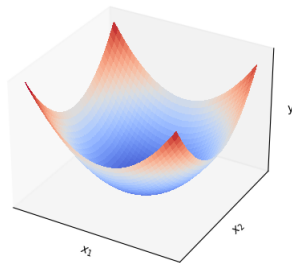
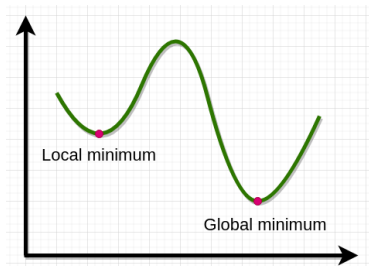
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Convex objective function

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- RSS is **convex**, which implies global minimum = local minimum.
- The left figure illustrates local minimum versus global minimum for a general function.
- The right figure is a classic example of bivariate convex function.



Minimizing the convex objective function

Find the global minimum of the convex objective function:

- Global minimum $\iff$ local minimum $\iff J'(\Theta^*) = 0$ and $J''(\Theta^*) > 0$.
- $\implies J(\Theta^*)$ is the global minimum iff

$$J'(\Theta^*) = 0, J''(\Theta^*) > 0$$

Minimizing the convex objective function

Analytic solution:

$$\begin{aligned} J(\Theta) &= \|\hat{f}_{\Theta}(\mathbf{X}) - Y\|_2^2 = (\mathbf{X}\Theta - Y)^T(\mathbf{X}\Theta - Y) \\ &= \Theta^T \mathbf{X}^T \mathbf{X} \Theta - Y^T \mathbf{X} \Theta - \Theta^T \mathbf{X}^T Y + Y^T Y \end{aligned}$$

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$$\frac{\partial^2 J(\Theta)}{\partial \Theta^2} = 2\mathbf{X}^T \mathbf{X} > 0 \text{ is for true.}$$

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$$\text{By (1), } \mathbf{X}^T \mathbf{X} \Theta = \mathbf{X}^T Y \implies \boxed{\hat{\Theta} = \Theta^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T Y}$$

See https://en.wikipedia.org/wiki/Matrix_calculus#Scalar-by-vector_identities for details on computing scalar-vector derivatives.



Part 3. Shrinkage methods

Expected prediction error

- The relation of input and output is modeled by the function f .
- Due to the noise from observation, $y = f(X) + \epsilon$, where $E(\epsilon) = 0$ and $\text{Var}(\epsilon) = \sigma^2$. Thus, the training data are always noisy.

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- For any fixed X with label y , the expected prediction error at X is

$$\text{EPE}(X) = E((y - \hat{f}(X))^2) = \boxed{\text{Bias}(\hat{f}(X))^2 + \text{Var}(\hat{f}(X))} + \sigma^2$$

$$\text{Bias}(\hat{f}(X)) = E(\hat{f}(X)) - f(X)$$

$$\text{Var}(\hat{f}(X)) = E((\hat{f}(X) - E[\hat{f}(X)])^2)$$

See https://en.wikipedia.org/wiki/Bias-variance_tradeoff#Derivation for the detailed derivation.

Bias-variance decomposition

The expected prediction error:

$$\text{EPE}(X) = \boxed{\text{Bias}(\hat{f}(X))^2 + \text{Var}(\hat{f}(X))} + \sigma^2$$

- The expected prediction error is composed of bias, variance and irreducible errors.

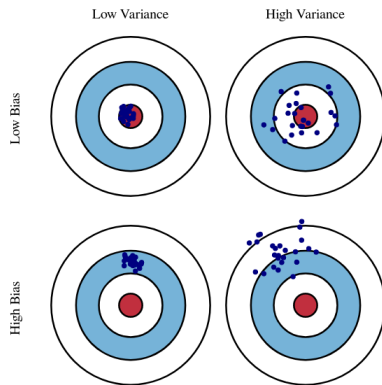


Image source: <http://scott.fortmann-roe.com/docs/BiasVariance.html>

Bias-variance decomposition

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- The expected prediction error is composed of bias, variance and irreducible errors.
- If simultaneously minimizing bias and variance, we can achieve a very considerable EPE.

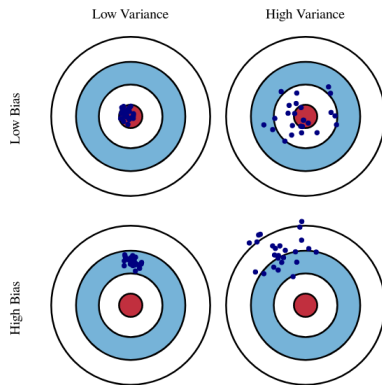


Image source: <http://scott.fortmann-roe.com/docs/BiasVariance.html>

Bias-variance tradeoff

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- Thus, we need to make a tradeoff between minimizing bias and minimizing variance.

Overfitting and underfitting

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Overfitting and underfitting

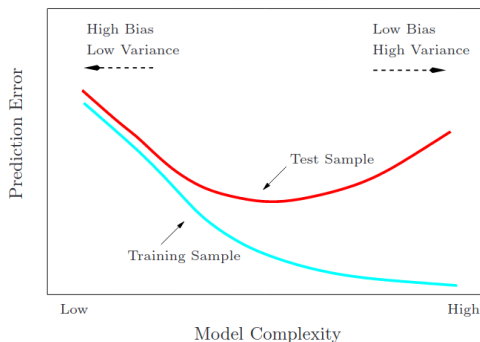
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Problems of ordinary least squares

- Ordinary least squares prone to learn a low bias, high variance model.
- To improve the EPE, sometimes we would like to sacrifice some bias to reduce the variance.
- Generally, tuning model complexity is a way to achieve better EPE.
- Shrinkage methods are batch of methods for automatically model complexity tuning, e.g., Ridge regression and Lasso regression.

Ridge regression

Recall the linear function in page 13: $\hat{Y} = \hat{f}_{\Theta, \theta_0}(\mathbf{X}) = \mathbf{X}\Theta + \theta_0$

- The ordinary least squares is an optimization problem:

$$\hat{\Theta} = \operatorname{argmin}_{\Theta} \|\mathbf{X}\Theta + \theta_0 - Y\|_2^2$$

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- Ridge regression aims to shrink the parameters by imposing a penalty on the square of the magnitude of the coefficients:

$$\hat{\Theta}^{ridge} = \operatorname{argmin}_{\Theta} \|\mathbf{X}\Theta + \theta_0 - Y\|_2^2 + \lambda \|\Theta\|_2^2$$

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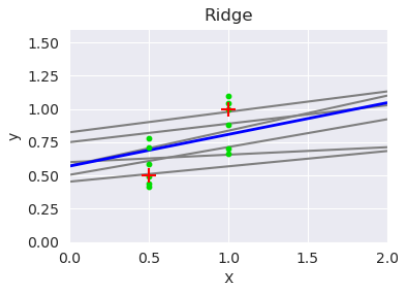
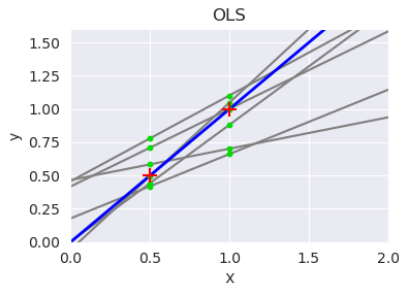
- The penalty term is $\|\Theta\|_2^2 = \sum_{i=1}^n \theta_i^2$ (a.k.a. L2 Norm)
- Note that θ_0 is not included in the penalty term.
- The λ is a hand-crafted hyper-parameter.

Understanding of the penalty term

- When there are many correlated features in a linear regression model, their coefficients can become poorly determined and exhibit high variance.
- A wildly large positive coefficient on one variable can be canceled by a similarly large negative coefficient on its correlated cousin.
- The penalty term regularizes the coefficients such that if the coefficients take large values, the optimization function is penalized. Therefore, the above problem is alleviated.
- The idea of penalizing L2 norm of the model parameters is also used in neural networks, where it is known as weight decay.

Ridge regression vs. ordinary least squares

- In the figures below, the red “+” markers are points on the true relation $f(X)$ (blue lines). We train linear regression models 6 times with noisy observed data at $X = 0.5$ and $X = 1.0$ (green points).
- Each time OLS is good at fitting training data but varies tremendously. On contrary, ridge regression suppresses the variation of models but fails to give a perfect fitting on training data.



Lasso regression

- Lasso regression is also a shrinkage method similar to ridge but with difference in the penalty term:

$$\hat{\Theta}^{lasso} = \operatorname{argmin}_{\Theta} \|\mathbf{X}\Theta + \theta_0 - Y\|_2^2 + \lambda \|\Theta\|_1$$

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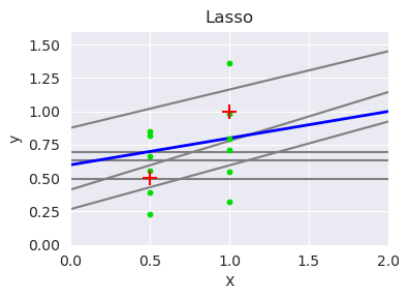
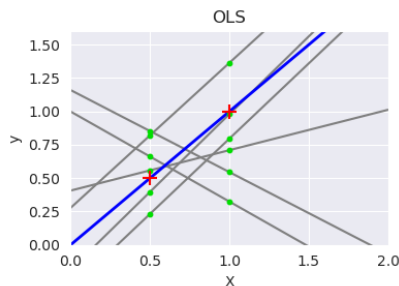
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- The penalty term is $\|\Theta\|_1 = \sum_{i=1}^n |\theta_i|$ (a.k.a. L1 Norm), which is designed for measuring the sparsity of parameters.
- Sparsity implies the number of zeros in Θ . If many elements in Θ are around zero, less features will be used, namely, the model complexity is reduced.

Lasso regression vs. ordinary least squares

- Conduct the same experiment in page 37.
- Similar to ridge regression, lasso regression yields models with less variation but is hard to give a perfect data point fitting.





Part 4. Summary

Summary

- Basic concepts and formulations of linear regression
- How to optimize linear regression RSS
- Bias-variance trade-off
- Ridge and lasso regression

Reading materials

- Chapter 3.1, 3.2, 3.4, 2.9, The Elements of Statistical Learning (2nd edition), T. Hastie, et al.
- Chapter 4.2, 4.3, Introduction to Linear Algebra (5th edition), G. Strang.



ESTR3108

Linear regression

- Understanding of the analytic solution
- Subspace perspective for linear regression

Understanding of the analytic solution

- Find the optimal solution of the convex objective function: $J(\Theta)$
- $J(\Theta^*)$ is the global minimum iff $J'(\Theta^*) = 0, J''(\Theta^*) > 0$
- Analytic solution:

$$\begin{aligned} J(\Theta) &= \|\hat{f}_{\Theta}(\mathbf{X}) - Y\|_2^2 = (\mathbf{X}\Theta - Y)^T(\mathbf{X}\Theta - Y) \\ &= \Theta^T \mathbf{X}^T \mathbf{X} \Theta - Y^T \mathbf{X} \Theta - \Theta^T \mathbf{X}^T Y + Y^T Y \end{aligned}$$

$$\begin{aligned} \frac{\partial J(\Theta)}{\partial \Theta} &= 2\mathbf{X}^T \mathbf{X} \Theta - \mathbf{X}^T Y - \mathbf{X}^T Y \\ &= 2\mathbf{X}^T (\mathbf{X} \Theta - Y) = 0 \dots\dots\dots (1) \end{aligned}$$

$$\frac{\partial^2 J(\Theta)}{\partial \Theta^2} = 2\mathbf{X}^T \mathbf{X} > 0 \text{ is for true.}$$

$$\text{By (1), } \mathbf{X}^T \mathbf{X} \Theta = \mathbf{X}^T Y \implies \boxed{\hat{\Theta} = \Theta^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T Y}$$

Questions on the analytic solution

Questions:

Given that $\Theta^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T Y$

- What if $(\mathbf{X}^T \mathbf{X})^{-1}$ does not exist, i.e., $\mathbf{X}^T \mathbf{X}$ is not invertible?
- Which condition is able to ensure $\mathbf{X}^T \mathbf{X}$ is invertible?

- Recall that

$$\mathbf{X} = \begin{pmatrix} \mathbf{1} & x_1^{(1)} & \dots & x_n^{(1)} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{1} & x_1^{(m)} & \dots & x_n^{(m)} \end{pmatrix} \in \mathbb{R}^{m \times (n+1)}$$

Discussion on $\mathbf{X}^T\mathbf{X}$

Theorem

$\mathbf{X}^T\mathbf{X}$ is invertible $\iff \mathbf{X}$ has linearly independent columns.

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Proof sketch:

- Firstly, note that $\mathbf{X}^T\mathbf{X} \in \mathbb{R}^{(n+1) \times (n+1)}$ (square matrix).
- When the columns of $\mathbf{X}$ are linearly independent, its nullspace only contains the zero vector, i.e., $N(\mathbf{X}) = 0$.
- So, we prove $\mathbf{X}^T\mathbf{X}$ and $\mathbf{X}$ have the same nullspace (a.k.a. kernel), denoted as $N(\mathbf{X}^T\mathbf{X}) = N(\mathbf{X})$.
- Then, $\mathbf{X}^T\mathbf{X}$, with the same nullspace $N(\mathbf{X}^T\mathbf{X}) = 0$, is invertible.

Discussion on $\mathbf{X}^T\mathbf{X}$

① Prove $N(\mathbf{X}) \subset N(\mathbf{X}^T\mathbf{X})$

$$\forall v \in N(\mathbf{X}), \mathbf{X}v = 0$$

$$\implies \mathbf{X}^T\mathbf{X}v = \mathbf{X}^T 0 = 0 \implies v \in N(\mathbf{X}^T\mathbf{X})$$

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- Therefore, $\mathbf{X}$ should be a “tall” matrix, i.e., $m \geq n + 1$.

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- Recall that $\mathbf{X} \in \mathbb{R}^{m \times (n+1)}$. So if $\text{rank}(\mathbf{X}) = n + 1$, $\mathbf{X}^T\mathbf{X}$ is invertible.
- Therefore, $\mathbf{X}$ should be a “tall” matrix, i.e., $m \geq n + 1$.
- Conclusion: if you want to solve a linear regression with n features, you need at least $n + 1$ independent training samples.

Subspace perspective

View linear regression problem from a subspace perspective.

- Linear equation: $Y = \mathbf{X}\Theta$, where $Y \in \mathbb{R}^m$, $\mathbf{X} \in \mathbb{R}^{m \times (n+1)}$, and $\Theta \in \mathbb{R}^{n+1}$.

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- The equation has solution(s) iff Y lies in the subspace S spanned by $\mathbf{X}$'s columns (i.e., feature dimension).
- If the equation has no solution, Y is off the subspace S .
 - Alternatively, we have to find the Y 's projection $\hat{Y}$ on S , such that $\hat{Y}$ is closest to Y among all other vectors in S .
 - We therefore solve $\hat{Y} = \mathbf{X}\hat{\Theta}$, formulate our goal by optimization:

$$\exists \hat{\Theta} \in \mathbb{R}^{n+1}, \quad \hat{\Theta} = \operatorname{argmin}_{\Theta} \|\mathbf{X}\Theta - Y\|_2, \quad \hat{Y} = \mathbf{X}\hat{\Theta}$$

Subspace perspective: a simple example

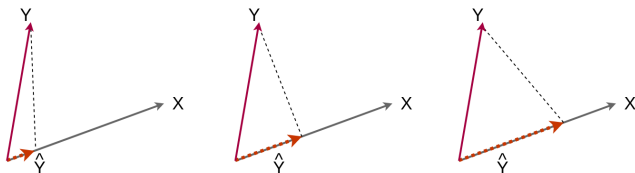
Take the univariate case as an example:

- Solve: $Y = X\theta_1$, where Y and X are vectors in $\mathbb{R}^m$ and θ_1 is a scalar. (Note that Y and X are vectors containing m observed samples, we omit θ_0 for simplicity.)

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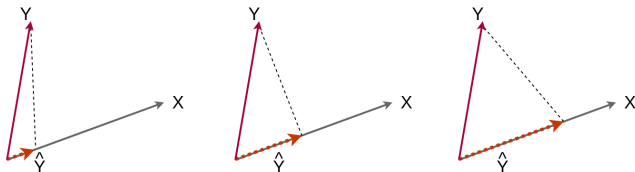
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- There exists a unique solution iff the direction of Y is parallel to the direction of X , i.e., Y lies in the subspace $\text{span}(X)$.



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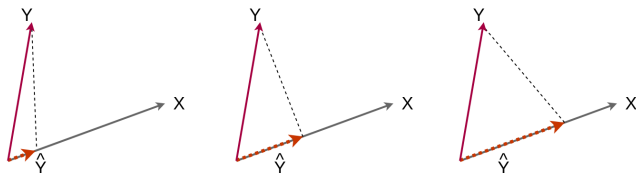
- Solve: $Y = X\theta_1$, where Y and X are vectors in $\mathbb{R}^m$ and θ_1 is a scalar. (Note that Y and X are vectors containing m observed samples, we omit θ_0 for simplicity.)
- If Y directs to a different way, in order to estimate $\hat{\theta}_1$, we need to find a vector $\hat{Y}$ that is parallel to X , and meanwhile has the closest Euclidean distance to Y .
- How can we determine the “best” projection $\hat{Y}$?



Subspace perspective: a simple example

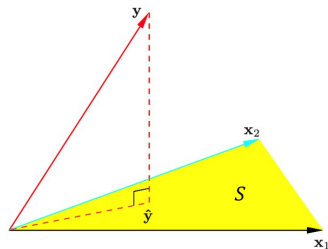
How can we determine the “best” projection $\hat{Y}$?

- In 2D Euclidean space, $\|\hat{Y} - Y\|_2$ is minimized when the difference $\hat{Y} - Y$ is orthogonal to X .
- We orthogonally project Y onto X and $\hat{Y}$ is the orthogonal projection.
- Finally, we have $\hat{Y} = \frac{X^T Y}{X^T X} X$, $\hat{\theta}_1 = \frac{X^T Y}{X^T X}$.



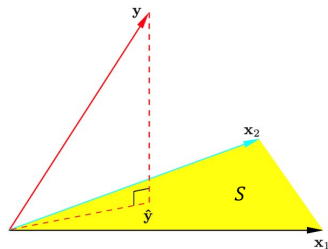
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Subspace perspective

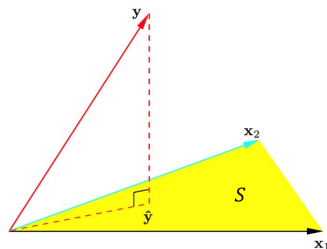
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- If a vector is orthogonal to $S \iff$ the vector is perpendicular to all columns (i.e., feature dimension) of $\mathbf{X}$, i.e., $\mathbf{X}^T v = 0 \iff v \perp S$.
- Thus, we have

$$\mathbf{X}^T(\hat{Y} - Y) = 0$$

$$\mathbf{X}^T(\mathbf{X}\hat{\Theta} - Y) = 0$$

$$\mathbf{X}^T\mathbf{X}\hat{\Theta} = \mathbf{X}^TY$$

$$\hat{\Theta} = \boxed{(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^TY}$$



Subspace perspectives

- The two methods for solving linear regression start from quite different perspectives. One is to utilize optimization techniques and one is to utilize the geometry understanding of linear equations.
- Nevertheless, they finally reach the same result.
(Note both derived in Euclidean metric.)